

Sequential Matching under Reciprocal Uncertainty: Ceilings, Acquisition, and Exact Matching

Round 1 Research Submission -- Vouchsafe Sequential Matching Hackathon

Team mohar-xe | Kit release 1.0.0 | 9 October 2026 | commit b1e1b46

Abstract

We study the Sequential Matching Hackathon: twice daily for 60 days (+40 follow-up), a policy spends a 12-unit ask budget and proposes disjoint feasible pairs to maximise mutual second-meeting intentions (MSMI) per 100 arrived members, averaged over six scenario families. Our contribution is measurement-first. Only 1.24% of pairs are hard-feasible (0.24% sparse); one third of members can never be matched; 98.5% of every introduction is lost to mechanisms no policy controls, above all a 3-day answer window discarding ~64% of successes. The only reliable pair signal is agreement on 4 of 7 soft fields. The reachable edge set caps introductions and greedy already realises 89-99% of the corrected ceiling -- so this is an acquisition problem, not an RL problem (honest headroom ~1.3-1.6x). Our stdlib-only policy climbs a paired ladder 0.333 -> 0.514 (directional; headline underpowered ~20x). Fixed-prior beats adaptive everywhere, as noise theory predicts.

1 Problem interpretation

Each of 60 days: (i) clarification asks costing ≤ 12 units (hard bundle = 3, one named soft field = 1; declined is permanent), then (ii) a batch of disjoint feasible pairs (empty = wait). Pairs must pass reciprocal hard checks both ways (age, gender, zones, smoking, children, structure, schedule), be unIntroduced, with all hard fields known. Feedback matures with delay; 40 follow-up days resolve outcomes. MSMI: both yes -> date within 30d -> both second-yes within 3d of the date. Score: mean over 6 families x 20 seeds. Tiebreakers: coverage, mutual acceptances, lower ask cost, lower inference time. JSON ≤ 1 MiB, 10 s/call, 2 cores, offline; memory self-carried. Soft fields are static (0/200 changed); only goal/pace/lifestyle/conversations enter the scorer (0.70 / 0.40 / 0.25 / 0.20; shifted: 0.25 / 0.80 / -0.25 / 0.50). The other three soft fields are dead weight: never ask about them.

2 Hypothesis

A reciprocal pair scorer learned online from delayed/censored feedback, with global max-weight allocation and VOI-ranked clarification, raises MSMI/100 over greedy via (i) learned weights vs uniform soft-count, (ii) global allocation vs greedy fill, (iii) soft acquisition vs none, (iv) drift-aware sequencing. Full RL is rejected (Learn2Match friction result: it fails exactly when volume is saturated and the gap is friction). LLM scoring is rejected (offline CPU; template conversations).

3 Reciprocal feasibility: introductions are capped

First-failing-rule on truth: who_to_meet 57%, age 26%, geography 12% (19% sparse); rest ~4% combined. $P(\text{matchable}) = 0.667 / 0.656 / 0.600$ -- one third can never be matched. $|E|$ is a hard ceiling (no repeats). A sparse contradiction ($|E| = 20.9$ vs 22.6 realised) was a seed-set mismatch: same seeds give $|E| = 22.8$ vs 22.6 (99% saturated), assigned subset of reachable with 0 violations. Clarification is supply-limited: 720 units vs ~80 askable members, exhausted by ~day 20.

Family	Match/200	$ E $	MaxMatch	Greedy
development*	133.4		103.1	34.5 ~92 (89%)
sparse	131.2	20.9-22.8	14.7	~99% (*)
cold_start	120.0		90.5	31.8 77.6 (86%)

* +delayed/shift/drift (structurally identical). (*) same-seed comparison.

4 Probability analysis (ranking only)

Per pair: one shared shock $s \sim N(0, 0.45^2)$; per direction accept $\sim \text{Bern}(\text{sigma}(-0.25 + \text{bias} + \text{fit} + s + \text{drift}))$, respond $\sim \text{Bern}(\text{rate})$; date w.p. 0.78; second yes $\text{sigma}(0.15 + \text{sbias} + s + 0.4[\text{goal match}])$. Directions correlate ($\rho \sim 0.43$): integrate over s (error $< 5e-6$), never multiply marginals. Replay validates ranking (top decile 0.041 pred / 0.033 emp vs bottom 0.002 / 0.000). Funnel (greedy, dev.): 91.6 -> 52.0 respond (57%) -> 12.6 mutual (24%, controllable) -> 10.3 dates -> 3.7 second-yes -> 1.3 in-window -> ~1.0 MSMI. The 3-day window alone destroys ~64%. Corrected joint (one E over s of all four sigmoids, both response stages): sum ~1.07-1.15 vs ~1.0 realised -- greedy at 87-93% of ceiling. Top/bottom-half lift 1.38x (fit) -> 1.48x (+oracle rate). DE weight search ruled out: range 0.132 < noise 0.356. Common random numbers cut SE 1.7x and are mandatory.

5 Proposed policy (implemented, stdlib-only)

- (a) Acquisition, matchability-gated: v1 wasted 183/384 soft asks (47.7%) on unmatchable members -- now gated. Zone-clustered hard
- clearance first; leftover on 4 live fields in tiers T0 (feasible pair)
- -> T1 (cleared) -> T2 (uncleared), by $|w_k|$. Adaptive bar.
- (b) Value model: observed: $+w_k$ if equal else $-w_k/2$; unknown:
- $E = w_k \cdot (1.5/|S_k| - 0.5)$ under uniform prior (exact; kills the
- None-equals-None artefact).
- (c) Weights: single-pass streaming logistic on matured responses (v1 re-fed events ~55x -- fixed); $\eta = 0.25/\sqrt{1+n}$, anchor 0.10
- to dev prior, clip 1.0. Synthetic check: pace 0.83 vs true 0.80.
- (d) Allocation: exact general-graph MWM (branch-and-bound, warm start):
- 268/268 fuzz-optimal vs brute force, < 1 ms/day. Graph is
- non-bipartite (~47% same-gender acceptance).

6 Missing and delayed data

Declined != not-asked (permanent, never bought). Missing responses are censored, never imputed; only matured outcomes update; pending queue by expected observation day. Assignment-day snapshots (no leakage; static fields remove staleness). No IPS (propensity null; pool splits only; same-seed simulator comparisons). Memory ~10 KB. Thompson sampling over pool priors; rank-only decisions; posterior-predictive coverage check as wrong-belief detector.

7 Evaluation

Baselines (identical seeds, all families): greedy, no-clarification, random-feasible. Ablations: no soft acquisition; fixed vs online weights; greedy-fill vs MWM; no drift scheduling; bandit off; theta sweep. Measured ladder (seeds 101-106, paired):

Configuration

	MSMI/100	vs greedy	Mutual
Organiser greedy	0.333	--	10.14
Candidate v1	0.347	+4.2%	10.56
VOI ask + greedy match	0.389	+16.7%	10.50
Full (VOI+MWM+adaptive)	0.444	+33.3%	10.81
Full (VOI+MWM+fixed)	0.514	+54.2%	11.20
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Honest reading: min detectable ~ 0.175 at $n = 36$, so only the fixed headline reaches it; per-variant cells ($n = 6$) are illustrative. Ladder is monotone, mutual +10%, and fixed beats adaptive even in shift (0.500 vs 0.417) -- as noise theory predicts (~ 50 trials cannot carry 4 weights). Cost note: we spend ~ 409 vs ~ 198 ask units (tiebreaker loss; Round 2: cost-capped ablation). Harness smoke test (seed 101, dev.): greedy msmi = 1 $\rightarrow$ team msmi = 3, both valid, 0.15 s/call.

8 Failure cases and limitations

- Withheld answers ($\sim 1/3$ unmatchable, kept in denominators); saturated
- sparse supply; 8-day-busy coupling (hence global matching);
- right-censoring (day-55 intros resolve $\sim$ day 77, unseen by policy).
- All outcomes synthetic -- no evidence about real relationships. Kit
- vendored verbatim (pin a8e26b3, MIT + synthetic-data licence); our
- work in research/; branch merged with history kept, policy moved out
- of the kit subtree. Reproduce: `unittest (22 OK) + verify_data.py`, then
- `analysis/{prob_model,validate_model,why_cap,ceiling,value_spread,`
- `candidate,verify_team_policy,team_policy_eval}.py`.

References

[1] Zong et al. Learn to Match. arXiv:2606.06744. [2] Li, Wang & Kong, Bandit Learning in Matching Markets survey, IJCAI 2025 (Liu-Mania-Jordan; Basu; Athanasopoulos et al. UAI 2026). [3] Udwani, non-adaptive greedy 1/2-competitive, arXiv:2403.18059. [4] Saar-Tsechansky et al., Active Feature-Value Acquisition, Mgmt Sci 2009; Ma et al., NeurIPS 2023. [5] Thompson (1933); Russo et al. (2018); Agarwal & Goyal (2013); Garivier & Moulines (2011); Joulani et al. (2013); Lakkaraju et al. (2017); Das & Kamenica (2005); Howard (1966). [6] Hayashi et al., OPE for matching markets, RecSys 2025. [7] Segal-Halevi (2011); Drummond & Boutilier, IJCAI 2013; Blum et al., OR 68(2) 2020.